19. The speed of a race car during the first 7 s of a race is given by:

t (s)	0	1	2	3	4	5	6	7
v (mi/h)	0	14	39	69	95	114	129	139

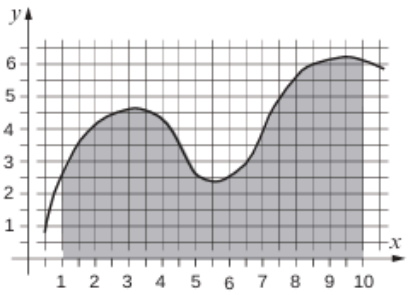
Determine the distance the car traveled during the first 7 s.

20. A rubber band is stretched by fixing one end pulling the other end. Measurements of the applied force at different displacements are given in the following table:

x (in.)	0	0.4	0.8	1.2	1.6	2.0	2.4	2.8
F (lb)	0	0.85	1.30	1.60	1.87	2.14	2.34	2.52

Determine the work done by the force while stretching the rubber band.

21. Use numerical integration to approximate the size of the shaded area shown in the figure. Create a vector with values of x from 1 through 10 and estimate the corresponding y coordinate. Then, determine the area by using MATLAB's built-in function `trapz`.



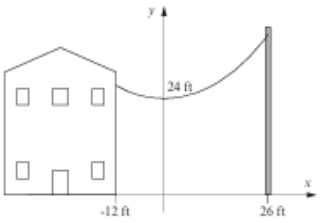
22. The electric wire that connects the house to the pole has the shape of a catenary:

$$y = a\cosh\left(\frac{x}{a}\right)$$

By using the equation:

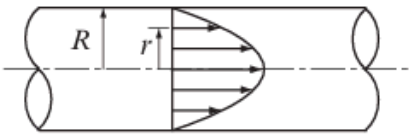
$$L = \int_a^b \sqrt{1 + [f'(x)]^2}\, dx$$

determine the length of the wire.



23. The flow rate Q (volume of fluid per second) in a round pipe can be calculated by:

$$Q = \int_0^r 2\pi v r\, dr$$



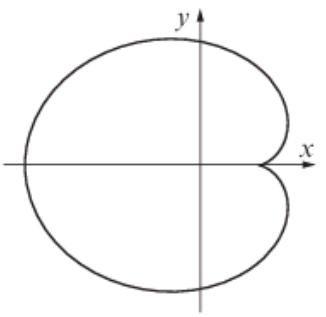
For turbulent flow the velocity profile can be estimated by: $v = v_{\max}\left(1 - \frac{r}{R}\right)^{1/n}$. Determine Q for $R = 0.25$ in., $n = 7$, $v_{max} = 80$ in./s.

24. The length of a curve given by a parametric equation $x(t)$, $y(t)$ is given by:

$$\int_a^b \sqrt{[x'(t)]^2 + [y'(t)]^2}\ dt$$

The cardioid curve shown in the figure is given by:

$x = 2b\cos t - b\cos 2t$ and $y = 2b\sin t - b\sin 2t$ with $0 \le t \le 2\pi$. Plot the cardioid with $b = 5$ and determine the length of the curve.



25. The variation of gravitational acceleration g with altitude y is given by:

$$g = \frac{R^2}{(R+y)^2} g_0$$

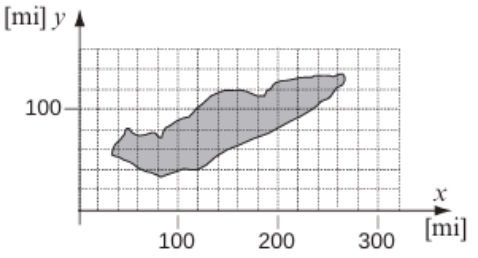
where $R = 6371\,\text{km}$ is the radius of the Earth, and $g_0 = 9.81\,\text{m/s}^2$ is the gravitational acceleration at sea level. The change in the gravitational poten-

tial energy, ΔU, of an object that is raised from the Earth is given by:

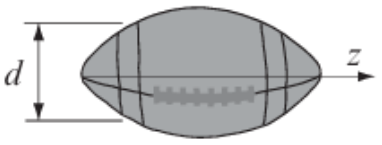
$$\Delta U = \int_0^h mg\,dy$$

Determine the change in the potential energy of a satellite with a mass of 500 kg that is raised from the surface of the Earth to a height of 800 km.

26. An approximate map of Lake Erie is shown in the figure. Use numerical integration to estimate the area of the lake. Make a list of the width of the lake as a function of x. Start with $x = 40\,\text{mi}$ and use increments of 20 mi, such that the last point is $x = 260$. Compare the result with the actual area Lake Erie, which is 9,940 square miles.



27. To estimate the surface area and volume of a football, the diameter of the ball is measured at different points along the ball. The surface area, S, and volume, V, can be determined by:



$$S = 2\pi\int_0^L r\,dz \quad \text{and} \quad V = \pi\int_0^L r^2\,dz$$

Use the data given in the table to determine the volume and surface area of the ball.

z (in.)	0	1.0	2.0	3.0	4.0	5.0	6.0	7.0	8.0	9.0	10.0	11.0	12.0
d (in.)	0	2.6	3.2	4.8	5.6	6	6.2	6.0	5.6	4.8	3.3	2.6	0